

1. World Time, Weather and Map

1a. World Time

Overview: The World Clock manages the passage of time. There are four distinct times of day, a seven-day week cycle, and a cyclical four-season year. The time of day, day of

week, and season will influence weather patterns, resource spawning rates, and creature behaviors and preferences.

Specifications:

1. Times of Day:

- The day is divided into four `time_of_day` segments: `AM Early`, `AM Late`, `PM Early`, `PM Late`. (These can be named something different; these just alphabetize nicely.) See Note at bottom about whether there should be any finer subdivisions of time, like “hour.” But for resource spawning and creature behaviors described in this document, nothing currently depends on any shorter time period than “Time of Day.”

2. Day, and Day of Week:

- Each `day` is represented by an incrementing integer (Day `0`, `1`, `2`, and so on).
- Days are sequentially ordered into Weekdays, with `day_of_week` progressing from `Monday` to `Sunday`. The game will continually cycle through this seven-day order.

3. Seasons:

- The game world has four longer-term “seasons” that are cycled through. There are multiple options for what this means, and how long the “seasons” last. This has implications for gameplay, data collection and art, so should be considered very thoughtfully AND tested with players before creating any art. Here are two examples:
 1. Ex. 1, Seasons: `Spring`, `Summer`, `Fall`, and `Winter`. Each season lasts for a duration of **28 days**, for a 112 game day full cycle. I use “Season” throughout the rest of the document, but note that this is malleable.
 2. Ex. 2, Moon Phases: `Waxing`, `Waning`, `Full`, `Crescent`. Each phase lasts for a duration of **7 game days**, for a 28 game day full cycle.
 3. Ex. 3: abstract World Stages, A through D. Each “stage” lasts until a “Disruption” event is triggered and shifts the players into the next “World Stage” where things now behave differently. (I like this one currently.)

Effects on Other Game Systems:

1. Resource Spawning:

- The probability of any given type of resource spawning will depend on the time of day (along with other things). Example: resources with “bug” type spawn behavior are more likely to spawn in the Early AM. See [Resource Spawning Patterns](#).

- Resource quality could be impacted by season, (e.g. "Strawberry Season"). To get highest quality resources for "Perfect Gift"-type goals requires players to be in the correct season. There is no current effect in place.
- 2. **Creature Behaviors:**
 - Creatures have different movement and activity patterns depending on time of day and day of the week. Some creatures are most likely to rest during "PM Late", while others might do an activity during "PM Early". See [Creature Behavior Patterns](#).
- 3. **Weather:**
 - The island weather patterns can be influenced by season, as described below.

Open Questions:

1. **How long should a 'cycle' be before repeating?** This question matters for seeing repeating patterns in data, and learning to "filter" data to a subset at the same time period. This depends on how much data would be collected in each time period before moving on. Each "season" above would last 28 days. Renaming this to "Moon Phases" (Waxing, Full, Waning, Crescent) would make shorter cycles, with each phase lasting only 7 days.
2. **Does art or lighting change based on time of day (or season)?** In the ideal case we'd have art and narrative distinguishing the seasons, plus noticeable effects on creature behaviors or spawning, etc. I think in the minimal case we could add the "season" variable, put something in the Newspaper, and mention things like "Rainy Season" etc (because there will be rain overlay, if not "Winter.")
3. **Should we use any finer grained time information (e.g. hours)?** This could matter for timed shop requests. It likely would not affect anything else though, like spawning or creature behavior.

1b. Weather System

Overview:

Weather will update daily on each island, with values depending on season. The Weather will affect creature patterns, resource spawning patterns, and/or creature item effects. Knowing the effect of weather on these variables should allow the player to adapt decisions (e.g. give different gifts on rainy days or forage in different areas), or to see patterns in the weather that let them predict what the weather will be (i.e. "forecasting"), to prepare (i.e. weather as a variable in "demand prediction").

Specifications:

1. The Weather is updated daily, and specifies the Sun and Rain conditions for each island. Despite being specific to each island, Weather can be considered to be a World property, if any world-level rules or patterns exist that make weather on each island dependent on weather on other islands. See Notes at bottom.
2. **Sun and Rain Conditions**
 - On any given day, the `sun_condition` on an island can be either `Sunny` or `Cloudy`. On any given day, the `rain_condition` can be either `Rainy` or `Dry`.

- These specific weather variables were chosen because they roughly align with the “Light/Shadow” and “Growth/Stability” balances, and can easily connect to faction stats or item boosts.
- The likelihood of these conditions varies with the seasons. See [Weather System code/config in Appendix](#). These probabilities can be adjusted. Currently:
 - **Spring**: 70% chance of rain, 70% chance of sun.
 - **Summer**: 30% chance of rain, 70% chance of sun.
 - **Fall**: 30% chance of rain, 30% chance of sun.
 - **Winter**: 70% chance of rain, 30% chance of sun.

Effects on Other Game Systems:

1. Resource Availability:

- The weather directly impacts the availability of resources on each island. Some resources might become scarce during rainy conditions, while others may be more likely to spawn when it is sunny. These are described in [Resource Spawning](#).

2. Creature Stats and Behaviors:

- [Creature mood or health](#) can depend on weather (Light creatures getting a boost on Sunny days, Shadow on Cloudy; Growth on Rainy days and Stability on Dry days).
- [Creature behaviors](#) (movements around the map, or item preferences) can change based on weather. For example, creatures could stay on their home island when it is Rainy, or could prefer ice creams on Sunny days. Currently there is no [Creature Behavior Effect](#) related to weather.

Notes and Open Questions:

1. Should weather be “magical” or real? “Chaos magic storms” or “rainfall”?
2. Does art or lighting change based on weather?
3. **Probable update**: The Sun and Rain base probabilities could also depend on island, in addition to season, so that e.g. Light island is more likely to be sunny than Shadow island.
4. **Probable update**: The weather could be made more predictable by adding a “Wind” value, of “None”, “North”, “South”, “East” or “West”. On the following day, the weather from one set of islands would be carried over to the adjacent islands, in the direction set by the Wind.
5. The Sun and Rain conditions could be decided at different levels, i.e. the Sun could be world-wide (same for all islands), while Rain is decided independently for each island.

1c. World Map

Overview: The game world is a nested hierarchy: World > Islands > Plots > Patches > Nodes. Each level in the hierarchy has associated level variables that are passed down to the lower levels, and can be used to determine spawn rate, type probabilities and qualities, and/or creature behaviors. At the lowest level, Nodes represent individual tiles where resources can spawn.

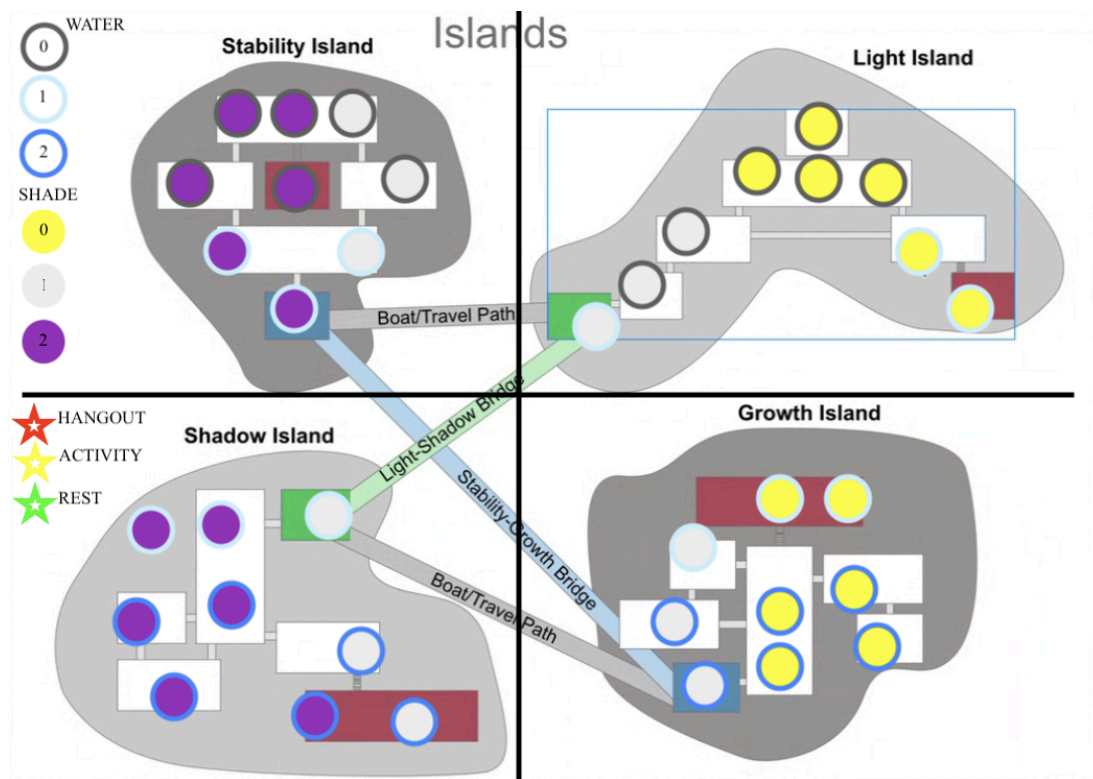
Hierarchy Levels:

1. World:

- The largest encompassing structure.
- Contains **Clock**, **Weather System** and **Islands**.

2. Island:

- There are 4 Islands in the World, each with a unique **name** (**Growth**, **Stability**, **Shadow** or **Light**).
- Each island has an **offset** for positioning. The four Islands are positioned roughly in the four “quadrants” of the world, with Light and Shadow diagonal from each other, and Growth and Stability on the other diagonal.
- Islands have an **island_health** variable, which is the aggregate of all its children **plot_health** variables.

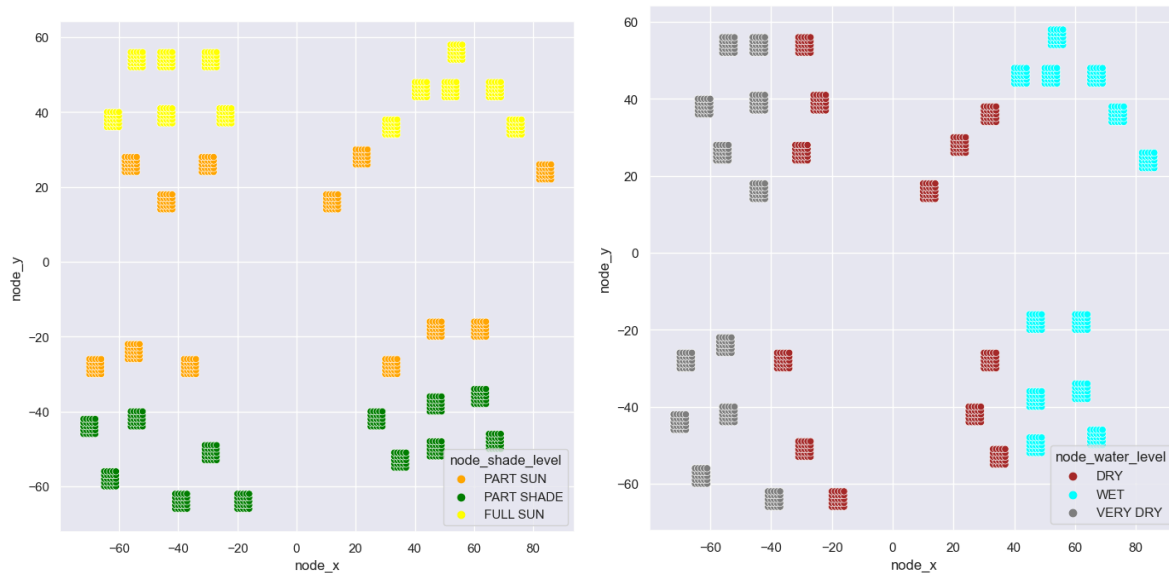


3. Plot:

- Each Island consists of 9 **Plots**. The 9 plots are (very roughly) laid out in a 3 x 3 grid, as in the current Growth Island map, so that they are spread out over the island.
- Plots can be thought of like “a clearing” or “a garden”, and are made up of smaller “patches” of resource spawn nodes.
 - More than one plot can appear on the same game screen, particularly if the game screen is large, as in the central screen on Growth Island. Any given “plot” should probably fit within one game screen, and not spill over or extend across more than one screen.
 - The patches and nodes contained in a plot should be close enough together that plot-level variables (currently, a “shade level” and “water level” as

described below) reasonably apply to all patches and spawn nodes contained in the plot.

- All plots should contain the same number of “patches,” and the same number of nodes. Currently a Plot contains 5 Patches of 5 Nodes each.
- Each Plot has a unique `plot_id`, `name` and `offset`. The `plot_id` should be a “Battleship” style identifier corresponding to its map location, like A1 (top left of map) or F6 (bottom right of map).
- Plots contain 5 smaller groups of tiles called `Patches`.
- The plot has a `plot_health` variable that is the aggregate sum over the `patch_health` of its children patches.
- The following are “stand-in” variables in the sense that they can be narrativized differently. But the nature of the variables and connections to other game systems should be maintained.
 - Plots each have a `shade_condition` variable that can be a number 0-3, mapped to `FULL SUN`, `PART SUN`, `PART SHADE`, `FULL SHADE`. The `shade_condition` for each plot is the plot’s default `plot_shade_level` + 1 if the island’s sun condition is Cloudy.
 - Plots each have a `water_level` variable that can be a number 0-100, mapped to `VERY DRY`, `DRY`, and `WET`, or `VERY WET`. This number can be affected by the weather, with the Rainy rain condition increasing water levels.
 - The water levels can change more slowly based on weather, by adding some constant amount (say, +10) to the plot’s water level when it rains (capped at 100). This number then decreases every Sunny day by -10 (floor at 0). The rate at which the water level decreases could depend on island, to create more dry conditions on some islands. Players will be able to adjust the water level of a plot, through a mechanism TBD. See Open Questions note at end of section.
 - All islands contain a mix of plots with different `plot_shade_level` variables, but the distribution varies. For example, Shadow island has more `PART SHADE` and `FULL SHADE` plots, while Light island has mostly `FULL SUN` and `PART SUN` plots. This distribution of plot variables will be used to generate the variation in type and quality of resources found at different map areas.



- A plot has an **anomaly** variable, hidden from the player, that tracks whether an anomaly is active on the plot (and possibly specifies an anomaly 'type'). TBD in [Anomalies](#).
- Plots (or entire screens) can be accessible or not to a player. Plots have an **access_threshold**, which is the number of friendships the player must have with the controlling faction to access the plot. Currently there is one [access controlled plot](#) on each island. Some special plots have higher probability of rare resources (see [Resource Spawning Patterns](#)).
- Plots each have **pastime_types** associated with different creature behaviors or activities. Plots may be places that creatures can **rest**, **hangout**, **tend** or do an **activity**. These variables are defined for each plot, but the logic is defined in the [Creature Behavior](#) section. Currently in the [Plot Variables List](#):
 - Each island has 5 **rest** plots. Creatures only rest on their own islands. So if a faction all rests at the same time, we'd expect to find ~5 on each 'rest' plot.
 - Each island has 4 **activity** plots, one per activity theme.
 - Each island has 5 **hangout** plots. Creatures can go to other islands to 'hangout,' choosing probabilistically according to preferences.
 - All island plots allow **tend**. Creatures go to any island to 'tend,' choosing probabilistically according to preferences.

4. Patch:

- Patches are contained within Plots, and consist of 5 individual **Nodes** placed near each other. These should be close enough that a "growth potion" can plausibly apply to an entire patch, and the patch would be expected to have similar growing conditions.
- Each patch has a **patch_id**. The **patch_id** should append a lowercase letter, 'a' through 'e', to the parent **plot_id**. Patch ids will look like: A1a, C2e, F6c and so on.

- Every Patch has a specific layout type, and an **offset** from its parent Plot.
- **Patch Health.** The state of a Patch is defined by its **patch_health**. The health of a patch is subdivided into four types, one for each faction. This can be thought of like "faction nutrients" at each patch location. See [Growth Potions and Island Health](#).
 - Patch health decreases when resources are foraged from that patch by a player, with the decrease applying to the faction of the resource. Foraging "Light" resources decreases the "Light" component of patch health. Each resource decreases the patch health by its **resource_value**.
 - Players can increase or alter **patch_health** by applying growth potions. I do think this could be potentially be interesting. "Patch health levels" are like 4 different parameters-- picture a little pie chart over each patch that can either have a dominant type, be balanced, etc. These can be affected by the growth potions. But resource spawning type/quality could depend on rules like "grows at full quality only when Light/Shadow are balanced" or "only when Growth is much higher than Stability" etc. So just this patch health variable and spawning dependence, plus the growth potion types, I think could open up some pretty neat little "gardening"-type data puzzles.
 - Patch health will update depending on creature stats (described in the section on [Update Logic](#).)

5. Node:

- Nodes represent a single game tile where a resource might be foraged. Five Nodes belong to each Patch.
- Each node has a **node_id**. The **node_id** should append a number, 1 through 5, to the parent **patch_id**. Node ids will look like: A1a1, C2e4, F6c2 and so on.
- A Node has an **offset** from its parent Patch, and a **coords** variable containing its x and y coordinates, which can be found by adding up all offset vectors from its parent patch, plot, and island.
- Nodes spawn resources that can be foraged by the player. The type and quality of resources spawned are probabilistic and can depend on any combination of nested variables passed down to the node. These relationships are defined in the section on [Resource Spawning Behaviors](#).
- A Node may have a node **type** that determines its visual appearance, user interaction, and resource spawn behavior. See Open Questions below.
 - Ex1 Node Type: "Overturned Rock" (example of a 'chancebox'). The player overturns a rock to see what resource may be under. There is a chance that there is no resource at all. If there is a resource found underneath, the resource may be more likely to have spawn behavior of 'bug' or 'rock' type.
 - Ex2 Node Type: "Flower Blossom" (example of a 'giftbox'). The player opens the petals of a flower to forage the resource contained inside. The player always receives a resource, and only the resource type varies. The resource may be more likely to have spawn behavior of 'plant' or 'mushroom' type.

Relationships across Map Hierarchy Levels:

- **Downward Effects:** Higher hierarchy levels pass variables down to their children. For example, the weather from the World and the Island faction can influence the type or quality of resources spawned at the Node level.
- **Upward Stats Aggregation:** Foraging resources at the Node level decreases the health of its parent Patch, contributing to a decrease in Island Health. “Balances” can be calculated at any level and affect spawn behaviors, creature behaviors, or trigger [Imbalance Events](#).

Relationship to Other Game Systems:

1. **Data System:** Nested variables affect the probabilities of spawning different types of resources at different locations within the game. The rules underlying the resource spawning appear as patterns in spawning or foraging logs.
2. **Environmental Control:** Players can influence variables at the Plot and Patch levels:
 - adjusting the `water_level` of a plot by adding water to a well or reservoir (TBD).
 - applying growth potions to alter a patch's `patch_health`. Selecting a patch and altering these variables allows players to improve their chances of getting desired resources.
3. **Foraging Effects:** The hierarchical structure allows for events or actions at one level to bubble up to affect higher levels. When players forage from a Node, this affects the Patch's `patch_health` and the island's aggregate `island_health`.

Open Questions:

1. **What are the two plot level variables? These will default to “water level” and “shade level” until there are any better ideas.** These can be “magic levels,” while “magic storms” happen to increase the “plot magic level.” But as illustrated by the water level and shade level variables, whatever these variables represent should: vary by plot, affect resource spawning on that plot, intersect with the “weather” system, update in time, and one should be directly adjustable by the player.
2. **How can players increase or decrease a plot’s “water level”?** One way: water level is represented by a “well”, “collection tank” or “reservoir”, and interacted with using a watering can. The watering can would add or remove water depending on whether it was full or empty, and players could move water from one tank to another. How players adjust the variable level will depend on the answer to Question 1.
3. **What are the Node types?** Whether nodes are ‘giftbox’ or ‘chancebox’ types matters for spawn logic, and weighting of resource types depending on `spawn_behavior`. See note in [Resource Spawning Behaviors](#) section.

2. Creatures and Factions

2a. The Creature Population and Factions

Factions Overview

The game has four factions of Creatures: **Light**, **Shadow**, **Growth**, and **Stability**. Each faction has a different visual appearance. Light creatures look like mushrooms; Shadow like bugs; Growth like trees; Stability like rocks.

Factions tend to share patterns of behaviors, respond to items in a similar way, and are more likely to be found in plots on their home island. These patterns are specified in [Creature Behavior Patterns](#).

Opposite, Synergistic and Neutral Faction Pairs

- The pairing of Light/Shadow or Growth/Stability are **Opposite Faction Pairs**. The pairing of Light/Growth and Shadow/Stability are **Synergistic**. Light/Stability and Shadow/Growth are **Neutral**.
- These faction pair relationships are used in **Crafting Logic** to determine the `faction_blend` of crafted items, which can influence the effects items have on creatures, and specifically the effect of [Growth Potions on Island Health](#).
- These relationships are also used in determining imbalances in gameplay, with imbalances in the relative faction stats of Light/Shadow and Growth/Stability determining probabilities of [Imbalance Events](#).

Faction Members

- Each faction has **25 Creatures**: 5 of each `creature_style` and 5 of each `creature_color`, for 25 unique Creatures per Faction (one of each unique color/type combination).
- The 5 Creature Colors are: Blue, Orange, Red, Green, and Purple.
- The 5 Creature Styles are: A, B, C, D, and E. These correspond to Body Style art variations for each Faction.
- Each creature has a unique `creature_id` and `creature_name`. **The `creature_id` is a number 0-99 (incremented in order by: Faction > Style > Color).**

Creature and Faction Stats and States

Each individual creature has **mood**, **health**, and **social** stats that evolve over time and in response to gameplay dynamics.

Mood, Health, and Social:

- The `mood`, `health` and `social` metrics are integer values between **1-100**, representing the creature's happiness, health and connectedness levels, respectively.
 - Note: The social stat is not the same as the creature's relationship level to each player, which is described below. The social stat will increase when the player gives the creature an item, and also when the creature goes to a "hangout" spot during the day. When the player gives an item to the creature, the social boost can also be applied to their relationship level. Ex: a creature is at 70 social. I give it a gift that boosts social by +20. The creature's social stat is now 90, and our relationship level is now 20. If I give it another +20 gift later, my relationship level to it is now 40, though its social stat has capped at 100. This way, the player's relationship level to any given creature can decrease if they don't pay attention to it for a long time, and they could lose "friendship" status, while the creature could still have a high social stat because other players are giving it many gifts.
- If a creature's mood, health or social stat is below a threshold, then they are "LOW MOOD" or "LOW HEALTH" and may alter their behaviors (e.g. go to the Medical Building instead of their normal pastime pattern. These effects will be described in the [Creature Behaviors](#) section.

Creature States:

- A creature can have states applied to it, which are labels that can affect its behavior in other contexts. Currently, these states will represent sicknesses and effects of anomalies. For example,
 - States are: Anomaly, Allergy, Sleeping Sickness, Flu.
 - **Allergy** is applied when creatures consume their "WORST" food item or potion. They lose health stat daily until removed.
 - **Anomaly** is applied when creatures consume a resource foraged from an anomalous plot.
 - **Sleeping Sickness** is applied randomly with some probability when creatures are LOW HEALTH. Affected creatures are 50% likely to sleep at any given time.
 - **Flu** is applied randomly to a creature, and has some chance of spreading to other creatures at the same Hangout area at the same time of day. They lose health and mood daily, and sleep more often.
 - If the creature visits a plot that `is_anomalous`, or consumes a resource that `is_anomalous` the creature can have the state `anomalous` applied to it along with a decrease in a stat. Anomalous states require a special potion to be removed, as described in the [Anomalies](#) section.
 - If the creature has a health value lower than some threshold value, the creature may have the state `Flu` or `Sleeping Sickness` applied to it. In the Medical building, the intake scan reveals the sickness directly, or e.g. `Cough` as a symptom. The state(s) of the creature, along with its faction, will determine potion effectiveness. See [Item Effects](#) section on potions for explanation.

- A state like `Allergy` could be applied to specific creatures who eat specific food items. The item would decrease their health and apply the state. As long as the state is applied they may lose health or mood daily.

Faction Stats

- Each faction has faction stats, comprising a `faction_health`, `faction_mood` and `faction_social` that are the sum over all of the member creatures' health, mood and social stats, respectively. These can also be reported as faction averages, by dividing the factions totals by 25 (creatures per faction).
- Other faction-level aggregate variables can count faction members with a specific state applied. E.g. "The Light faction has 12 creatures with `Cough`." These aggregate variables, e.g. "faction cough count" may be revealed in the [Newspaper](#) as "headlines".

Between-faction Imbalances in Stats

- If two factions, particularly 'opposite' factions, have a relative difference in stats above a threshold value, this is an 'imbalance'. I mention a way of calculating this in the section [here](#).
- Imbalances can be between factions on a specific stat (e.g. "health disparities"), or an "overall" imbalance across totals (all health, mood and social combined). An Imbalance should be thought of at the Population or World level. Whether or not an imbalance exists at this level could be used later to determine spawning or creature behaviors (e.g. Light creatures will no longer visit Shadow island if there is a Light/Shadow imbalance).
- Imbalances may trigger an [Imbalance Event](#). The simplest version of this for now is to set a threshold difference between two faction stats, e.g. between the `faction_health` values for Light and Shadow factions.
 - Moderate imbalances can trigger a Newspaper "headline" alerting the players to an imbalance. Ex: "Stability creatures thriving, while Growth creatures remain sick." See: [Newspaper](#) section.

Player Relationships to Creatures and Factions

- **Player-Creature Relationship Levels:**
 - Creatures have a relationship level with each individual player in the game, stored as an integer ranging from **1-100**.
 - The relationship level for a given player tracks the effects of items given by that player to the creature's social stat. Example: if a player gifts an item with a +20 social stat boost, the player's relationship level with that creature increases by +20, even if the creature's social stat is already maxed out.
- **Player-Creature Friendship:**
 - If a creature's relationship level with a specific player is above a threshold value, the player and creature are considered **"friends"**.
 - Each creature has a `friendship_threshold`, making it more or less challenging to befriend some creatures or creature types. Currently this threshold is associated

with the `creature_style`, e.g. "Growth Style A creatures are very friendly; Growth Style E are hard to befriend."

- **Player-Faction Friendship:**
 - If a player has a threshold number of friendships with creatures within a faction, then the player is "Faction Friends" with that faction. A reasonable threshold is probably 13 creature friends (more than half the faction). This threshold could depend on faction, or be the same for all factions.
- **Human-Faction Friendship:**
 - If a threshold number or fraction of players are "Faction Friends" with a faction, that Faction is "Friends" with Humans. A reasonable threshold is "more than half the players are faction friends."
 - All Four factions becoming "Friends with Humans" can be a game completion goal.

Connections to Other Game Systems

- **Altering Creature Stats with Items:**
 - Creature stats can be altered by items such as **potions**, **gifts**, or **foods**. The effects of items on creatures are described in the [Item Effects](#) section.
- **Connections between Stats:**
 - The creature stats might increase or decrease based on other stats, Faction stats, Island Health, etc. For example, if creatures are low health they may also start to decrease mood. These relationships will be specified in the section on **Update Logic**.
- **Map Access and Creature Friendships:**
 - Players may gain access to a special map plot by becoming friends with a threshold number of creatures within a faction.
- **(Possible) Creature Stats influence Item Effects:**
 - Creature stats can influence their "preferences" for specific items (can modify item effects). Ex: a creature with a low health stat may not have their mood improved by gifts until their health improves. See the **Item Effects** section.

2b. Creatures List

See [Creatures List](#) in Appendix.

2c. Creature Behaviors

CB1: Creatures Choose Pastimes, and Move to Plots

Each time of day, each creature chooses a 'pastime' from: Activity, Hangout, Rest and Tend. The selected 'pastime' determines where that creature is likely to be found on the map (in which screen, or plot), and which of their stats should update. Creatures move to the chosen plot, though players

can “override” by interacting with creatures and sending them elsewhere (for example, to the Medical Building).

- Factions tend to have different time of day patterns. For example, Light creatures tend to Rest in "AM Early" and Growth Creatures in "PM Late." These are specified in [Creature Behavior Patterns](#).
- Creatures do not need to be animated “playing football” or “resting” or to visually appear different in any way for this system to work. Visual effects can be added if scope permits. The player just needs to be able to predict where the creature is likely to be found at some particular time of day, knowing which faction, type and color it is, and possibly knowing about the creature’s preferences based on their shop requests or gift item reactions, which all rely on the same preference logic as specified in [Creature Behavior Patterns](#).
- Pastimes determine which map plots the creature can move to. Each of the 9 plots on each island is tagged as allowing or not allowing specific pastimes and/or activity types:
 - For Activities, each activity type has one possible plot (determined by activity choice described below). (4 'activity' plots per island.)
 - For Rest, creatures stay on their home island on one of multiple plots (5 'rest' plots per island)
 - For Hangout, creatures select from 'hangout' plots according to island preferences (4 'hangout' plots per island)
 - For Tend, creatures select from 'tend' plots according to their island preferences. Currently, all plots are plots where creatures can ‘tend’, so there are 9 'tend' plots per island, and the creature can be found at any of them, though the probabilities are weighted as specified by their island preference in [Creature Behavior Effects](#)). When creatures ‘tend’ at a plot, their stats are lowered while the health of a patch is increased.
 - What’s happening could be narrativized in various ways. The creature could be “tending to a garden”, “planting seeds”, “doing creature magic” or “putting a spell on a plot”, “working the land”, or “pooping on a patch” and fertilizing it. It doesn’t really matter for current purposes. The result is the patch is more likely to spawn that faction’s resources if the creature does it, and when the creatures have any low stats they’re less likely to do it, so island health drops, their resources are more scarce, and all this can easily generate data since it happens by an actual logic/mechanism.
- Activity gives a small boost to creature mood, Hangout to social, Rest to health. Tend decreases creature stats, but increases patch health.
- Item effects are altered and/or boosted during each pastime. Giving a creature a Food while they are at a ‘hangout’ can have an enhanced effect (and maybe greater consequence for ‘incorrect’ food). Giving a creature a health potion while they are ‘resting’ can have an enhanced effect on health, but may lower their mood.
- If creatures have “LOW MOOD”, “LOW HEALTH” or “LOW SOCIAL” (below some defined threshold) they will not select a pastime as normal. Instead they (TBD):
 - LOW HEALTH: go to Medical Center or Rest.
 - LOW MOOD or LOW SOCIAL: only do Activity/Hangout until Mood/Social above some other (higher) threshold stat.

- If a creature has Sleeping Sickness applied, they have a 50% chance of sleeping at any time of day.
- Players could potentially override the creature's pastime choice, and be able to send a creature to a particular activity, plot or hangout, as they currently do to send them to the Medical Building. This is raised in the Open Questions section below.

CB2: For "Activity" Pastime, Creatures Choose an Activity Theme and Type

When creatures do an "activity" pastime, they choose a specific activity theme and type according to their preferences (see [Creature Behavior Patterns](#)). This choice will determine which map plot they go to, and may add boosts to the effects of items gifted to them at that time and plot location.

Themes and types will be decided through co-design with playtesters. Currently as stand-ins, there are four Activity themes, each with four types. These were designed based on faction/game themes, to fit with existing resources and crafting constraints, connect to other game systems like bridges, player custom items, and watchers:

- **Music/Harmonics** (preferred by Growth)
- **Building/Sculpting/Stone Masonry** (preferred by Stability)
- **Reading/Writing** (scholarly/secretive pursuits; preferred by Shadow)
- **Fabrics/Fibercrafts** (clothing, fashion, crafting; preferred by Light)

How are these activities shown in-game? Minimal case/For the time being, activities could be just things that get logged, and/or if you interact with a creature the creature text says like "Current activity:{creature.activity}." Otherwise they just appear in the activity-relevant plots.

CB3: Creatures Make Shop Requests

Creatures send requests to the shop for resources and items of different categories, types or qualities, and offer trades. The requests made by each creature depend on their Activity Preferences described above, so that requests logs provide information about creature behaviors (and where they're most likely to be found during Activities), and vice versa. Danny is currently working on this system.

Q: Do creatures literally travel to the board and "make" a request? Does the player see this happening? No, these can be generated "events" and can reuse the creature "preferences" logic. The actual corresponding creature should continue walking around the map as normal. "Shop requests" can be made at random by some separate system, or at specific times (e.g. "rush hour on Saturday"), or could be triggered by a creature when a stat is low. The requests can be generated by re-using the creature activity weightings and "preferences" logic that specifies which creatures like which activities (if they like sports 3x more than art, they're 3x more likely to request a sports item, etc).

Connections between Creature Behaviors and Creature Stats/States:

- The "Rest" pastime increases the health stat of the creature. "Activity" increases the mood stat. "Hangout" increases the social stat. "Tend" decreases all stats, but increases a random `patch_health` on the selected plot.
- If a creature's mood, health or social stat is below a threshold value, they are "too sick" or "too lonely" to do their activities, and may instead visit the Health Center, and/or *only* Rest until they are above some threshold value again.
- Anomalies can affect creature stats and state: its health (and/or mood) stats can be decreased, and a state applied (e.g. "Cough", "Sick", "Anomaly Type A"). Applied states can update creature stats (e.g. health decreases by 2 points every day the creature has "Cough" applied), or can alter behavior patterns (if "Anomaly A" applied, creature only chooses Activity, or now prefers the opposite island). These will be specified in a section on [Anomalies](#).

Notes and Open Questions:

1. The [Anomalies](#) logic is TBD.
2. The "pastime" overrides based on creature stats is TBD. If low health it makes sense that the creature goes to the Medical Center. What if they're low mood? Low social? Should low health prevent them from improving mood? Vice versa? Owen notes: "Creatures going to the medical center on their own is cool, but easily could get out of hand and EVERYONE goes to the med center."
3. The player can currently tell creatures to go to the Medical Building. Can the player tell the creature to "Go play Football" if they notice their mood is low? Or "go rest", "go to Plot 5 and hang out" etc?

2c. Creature Behavior Patterns

This list will be modified and extended. The effect details are currently specified in [Creature Behavior Effects](#) in the Appendix.

- **CBP 1:** Factions choose different pastimes at different times of day.
- **CBP 2:** Factions have different preferences for activity themes.
- **CBP 3:** Creature styles have different preferences for activity themes.
- **CBP 4:** Creature colors have different preferences for specific activities.

3. Resources and Spawning

3a. Resources

Overview

Players can collect 16 different types of resources. All resources spawn across all islands. Each faction has a unique resource for each of the following categories:

1. Fruit
2. Sweet
3. Magic
4. Material

These resource categories are combined to craft different categories of item: Foods, Gifts, Health Potions and Growth Potions. The specific item type crafted depends on the specific resource types used, while the quality of the item will depend on the qualities of the resources used.

Resource Type Properties

Each resource type has a:

- **Name:** Unique text representing the resource name, e.g. "Stonefruit."
- **resource_type_id:** a lowercase letter representing the spawn type, plus a number 1-4. Resource type ids then look like: p1, p2, p3, p4 for plant 1-4, and b1, b2 for bugs 1-4.
- **Faction:** The faction to which the resource belongs (Growth, Stability, Light, Shadow). This variable currently only relates to the patch health depleted when the resource is foraged, and to the Growth Potion crafting logic and effects.
- **Category:** One of four categories of resource (Fruit, Sweet, Magic, Material). These categories are used in crafting logic.
- **Commonness:** A numerical value representing the relative probability of that resource spawning. Higher numbers indicate greater probability.
- **Base Value:** The resource type's 'base value' is the decrease to `patch_health` when this resource is foraged from a patch. Is inversely proportional to commonness. Higher value resources create higher value items, which have larger impact on creature stats when gifted or used.
- **Quality:** A number 0-100 representing a "percent" quality, or 'effectiveness'. The resource value times its quality is the total possible contribution to crafted item value, so items crafted from lower quality resources will have a lesser effect on creature stats. Note: this means that foraging resources of low quality causes larger decreases to island health than boosts to creature stats.
- **Spawn Behavior:** Currently, resources may have one of four "spawn behaviors": plant, bug, mushroom, rock. These are not necessarily tied to any faction. Spawn behaviors can be used to modify spawn probabilities based on variables like `time_of_day` (e.g. "bugs come out in

the morning"), or based on spawn node types if there are multiple (e.g. "bugs are found by overturning rocks").

- **Shade and Water Preferences:** Additional resource variables specify how resources interact with map variables in determining spawning probabilities. Currently, `shade_preference` is specific to each resource at the individual level, and can be used in the spawning logic to weight the types or adjust qualities of resources spawned.

Spawned Resource Instances

When spawned, each resource has:

- **Unique ID:** Unique id representing the specific resource. Used for tracking resources. Begins with the id of the resource type (e.g. "p1-" or "b3-") and then appends an increasing integer.
- **Spawned Resource Type:** One of the 16 resource types, e.g. "Stonefruit."
- **Spawned Resource Value:** Resource value times the resource quality (as determined at spawning). Rare (low `commonness`) resource types have higher value, so have higher max `resource_quality`.

List of Resource Types and Properties

See [Resource Types and Properties](#) list in Appendix.

Considerations and Balancing

- The resource commonness and value parameters, and effects of world variables on spawning, will all need to be tuned in response to e.g. player foraging rate and how likely players are to "overforage", how many gifts/potions it should require to raise a creature's stats to 100, how easy it should be to trigger an imbalance, etc.
- These resource commonness values are currently balanced to add up to the same number (16) for each faction, though weighted so that different factions produce more of a particular resource category.
 - *Shadow* faction makes more *Magic* resources.
 - *Light* faction makes more *Sweet* resources.
 - *Growth* faction makes more *Fruit* resources.
 - *Stability* faction makes more *Material* resources.
- These values are also weighted to have a ratio of 3 : 2 : 2 : 1 for magic : fruit : sweet : material, reflecting the ratios in which they appear in recipes outlined in the **Crafting Logic** section.

3b. Resource Spawning Behaviors

Spawn Rates, Locations, Type and Quality

- Spawn rate is currently determined by the total patch health at each node, proportional to `patch_health/PATCH_MAX`. As a patch is depleted, it will be less likely to spawn anything. At low enough patch health, higher value resources can not spawn at all, because they exceed the current patch health.
- These are all related questions about spawn rate, depending on how nodes spawn, and require probably one design/development conversation between Owen, Lisa and Danny: whether they have “types” (“giftbox” or “chancebox”), whether spawning happens at set time intervals or continuously and probabilistically, whether a node waits and “recharges” before spawning again after being foraged, and (if “giftbox”) how long spawned resources stay available before “decaying” and resetting the node.

Base probabilities of finding each resource

- There are 16 resources, each with a base "commonness" reflecting the relative probability that it should be spawned. These commonness values are the weights in a random choices function.
- These weights are modified by "mods" determined by variables at each level of the map hierarchy, to adjust probabilities of resource types spawning in response to e.g. rain or time of day. These mod functions are listed in the section on [Resource Spawning Patterns](#).
- The 'mods' determine various resource spawning patterns as described below, e.g. "Resources with different spawn behaviors tend to spawn at different types of day."

3c. Resource Spawning Patterns

The following effects on spawning determine overall patterns in resource type and quality. Specific multipliers used are in the [Resource Spawning Patterns](#) section of the Appendix.

RSP 1. Different resource types spawn at different times of day, depending on their `spawn_behavior` (if the resources behave like bugs, plants, rocks or mushrooms). Currently, for spawn behaviors: “bug” more likely in AM Early; “rock” more likely in AM Late; “plant” more likely in PM Early; “mushroom” more likely in PM Late.

RSP 2. Different resource types are more likely to spawn in the Rain or Sun, depending on their `spawn_behavior`. Currently, ‘bug’ and ‘mushroom’ are more likely when Rainy; ‘plant’ and ‘rock’ more likely when Sunny.

RSP 3. Different resources grow best in different shade and water conditions, depending on the `shade_preference` and `water_preference` of the resource type.

RSP 4. Patch health levels affect spawn rates, resource type and resource quality. Because creatures of each faction contribute their particular type of ‘energy’ to `patch_health`, this has the effect of faction-specific resources being spawned preferentially on each island. The difference between doing it this way vs. establishing a general rule like “faction-specific resources are always

more common on their own island” is that now when the creatures of a faction are sick, those faction resources are less common. The benefit of doing this by an actual underlying mechanism, rather than narratively or by threshold, is that the processes involved (tending to plots) can all, in principle, be recorded and observed in data.

RSP 5. Some resources grow preferentially in specific plots. Some resources (specifically less common ones) can grow preferentially in a specific plot, and most likely the one that has a `player_access_threshold` set.

RSP 6. Node x, y coordinates affect resource quality for some resources. I didn’t implement this one yet.

Others: We could add:

- Resource base rates or quality depend on season.
- Resources require different patch health levels or combinations to spawn. E.g. some resources only spawn in patches with balanced Light/Shadow, or High Light/Low Shadow, etc.

Notes and Open Questions:

1. The specific patterns are all still being designed and are subject to change. The general idea that variables at various levels of the island hierarchy, and/or creature stats, affect spawning rates, types and qualities in various ways, is unlikely to change.
2. It could be that the spawn rate at a patch is related to the overall island health, or plot health, and not to the specific patch health.
3. Knowing where resources of specific types and qualities are most likely to spawn is only effective to the extent that finding specific types or qualities of resources is motivated within the game, and the differences in probabilities and/or qualities is significant. For example, if there’s a 10% boost on finding a resource across the map, there’s very little reason to travel to boost chances by that amount. Some considerations:
 - a. Resources should be overall probabilistic but with strong effects on probability of finding them in specific areas.
 - b. The quality effect can also be large. Quality may not matter initially because players can craft low quality items with low quality resources.
 - c. These patterns can depend on time. For example, if the patterns shift more substantially by season, the player could look back at a previous season’s data for clues about where resources are likely to be found.

3d. Resource Spawning “Overrides”

The above rules and patterns create a default probability of resources spawning with a particular type and quality. Overall, they currently only affect the probability of finding a resource (even if the probability effects are strong). Adding “overrides” to probabilistic spawning could enable pattern detection and prediction of resource spawning in various ways:

- **“Traveling” Spawners.** Creating an underlying pattern in some resource spawn behaviors. For example, a particular resource with “bug” spawn behavior could select a new patch or plot to “override” per day. Instead of that patch/plot spawning according to the default probabilities, it would be overridden by a different probability distribution in which that “bug” type resource was very likely.
- **Growth Potion Combos.** Adding growth potion combos or other upgrades that allow a player to determine what will spawn the next day. This could function as a single-time probability modifier that multiplies all other resources by 0. Continuing the 4x4 matrix model used in crafting, GP1 and GP2 inputs could select the row and column of the resource matrix, to apply a one-time deterministic “override” to guarantee that resource.
- **Imbalance Events** can trigger all nodes on an island to spawn a type of resource from the overpowered faction. For instance, if Light/Shadow is imbalanced in favor of Light, then all nodes on Light island can spawn Light resources. Because foraging depletes that resource from the island, this would allow players to forage resources to address the imbalance (addressing the “foraging isn’t always bad” requirement).

4. Recipes and Crafting

4a. Crafting Overview

Players can craft four categories of items. Each category of items has: 1 of 4 themes, and 1 of 4 item types, for 16 unique recipes per category. These are listed in the [Items List](#) in the Appendix. The item categories are:

1. Foods
2. Health Potions
3. Gifts
4. Growth Potions

Effects. Foods, gifts and health potions will have different effects on creature health, mood and social, while growth potions will affect patch health (as described in the section on **Item Effects**).

Crafting. These items are crafted by combining two input resources, via the logic described in the following section. Crafted Item instances have:

- a unique **item_id**: Could be a string representing the item type then unique incrementing number, like: hp1a-93, g3b-0. Otherwise could be made by appending the modifier resource id to the base resource id: b1-167-p3-99. The first would hint at the type, the second at the resources used to craft it.
- a category (Food, Gift, Health Potion, Growth Potion)
- a theme (e.g. "Music", "Cupcakes")
- a type (e.g. "Violin", "Funfetti Cupcake")
- a faction (Light, Shadow, Stability, Growth)
- a faction_blend (Specific, Balanced, Synergistic, Neutral)
- Suggestion - Change Result to Reaction(sum of value1*quality1 + value2*quality2 of the ingredient resources)
- an item quality (can come from crafting success/skill level, or default to 100%)

Open Questions:

1. Are which items that can be crafted be constrained by context (e.g. in the Medical Building, the player can only craft Potions)? Owen: "I think we wanted it to be this way - like you can only brew potions in the medical center. Maybe there's a workbench somewhere that allows you to craft."

4b. Item Categories and Themes

Foods

There are 16 Food items belonging to 4 Themes: **Smoothie**, **Jelly**, **Pastry**, **Cupcake**.

Foods are crafted from combinations of a **base** **sweet** resource and **modifying fruit** resource. The base sweet resource chooses the theme, and the modifying fruit resource chooses one of the four themed types.

Base 'sweet' resources:

- *Nectar* (plant). Makes: *Smoothie*.
- *Sugar Crystals* (rock). Makes: *Jelly*.
- *Honeycomb* (bug), Makes: *Pastry*.
- *Powdered Sugar* (mushroom). Makes: *Cupcake*.

Modifier 'fruit' resources:

- *Wheatberry* (plant). Flavors with *Berry*.
- *Stonefruit* (rock). Flavors with *Peach*.
- *Cocoaberry* (bug). Flavors with *Chocolate*.
- *Sporefruit* (mushroom). Flavors with *Tropical*.

1. Base/Theme		2. Modifier/Type			
		Wheatberry	Stonefruit	Cocoaberry	Sporefruit
Nectar	Smoothie:	<i>Berry Smoothie</i>	<i>Peach Smoothie</i>	<i>Chocolate Smoothie</i>	<i>Tropical Smoothie</i>
Sugar Crystals	Jelly:	<i>Berry Jelly</i>	<i>Peach Jelly</i>	<i>Chocolate Jelly</i>	<i>Tropical Jelly</i>
Honeycomb	Pastry:	<i>Berry Pastry</i>	<i>Peach Pastry</i>	<i>Chocolate Pastry</i>	<i>Tropical Pastry</i>
Powdered Sugar	Cupcake:	<i>Berry Cupcake</i>	<i>Peach Cupcake</i>	<i>Chocolate Cupcake</i>	<i>Tropical Cupcake</i>

Health Potions

There are 16 Health Potions belonging to 4 Themes: **Rejuvenation Tea**, **Protection Tonic**, **Enchantment Brew**, **Luminous Bubble Tea**.

Health Potions are crafted from combinations of a **base magic** resource and **modifying fruit** resource. The base magic resource chooses the theme, and the modifying fruit resource chooses one of the four themed types.

Base 'magic' resources:

- *Vitalvine* (plant). Makes: *Rejuvenation Tea*.

- *Mycolumin Essence* (mushroom). Makes: *Luminous Bubble Tea*.
- *Enchantwings* (bug), Makes: *Enchantment Brew*.
- *Vitalite Crystal* (rock). Makes: *Protection Tonic*.

Modifier ‘fruit’ resources:

- *Wheatberry* (plant). Flavors with *Berry*, *Wheatberry* or *Plain/Vanilla*.
- *Stonefruit* (rock). Flavors with *Stonefruit* (e.g. peach, apricot).
- *Cocoaberry* (bug). Flavors with *Chocolate*.
- *Sporefruit* (mushroom). Flavors with something *Tropical* or *Fruity*.

1. Base/Theme		2. Modifier/Type			
		Wheatberry	Stonefruit	Cocoaberry	Sporefruit
Vitalvine	Rejuvenation Tea:	Wheatberry Green Tea	Peach Oolong Tea	Cocoa Mint Tea	Tropical Tea
Vitalite Crystal	Protection Tonic:	Wheatberry Guard Tonic	Stonefruit Shield Syrup	Cocoa Barrier Blend	Sporefruit Defense Tonic
Enchantwings	Enchantment Brew:	Wheatberry Luminescent Brew	Stonefruit Illusion Brew	Cocoa Mystic Brew	Sporefruit Enigma Brew
Mycolumin Essence	Luminous Bubble Tea:	Wheatberry Glow Boba	Stonefruit Crystal Boba	Cocoberry Bliss Boba	Sporefruit Sparkle Boba

Gifts

There are 16 Gifts belonging to 4 Themes: **Music, Building, Reading, Crafts**. (Except I rename them below to **Harmonics, Masonry, Secretspinning**, and **Mycofibercraft**, respectively.)

Gifts are crafted from combinations of a **base material** resource and a **modifying magic** resource. The base material chooses the theme, and the modifying material chooses one of the four themed item types.

Base ‘material’ resources:

- *Heartwood* (plant)
- *Moonstone* (rock)
- *Dark Silk* (bug)
- *Fungus Fiber* (mushroom)

Modifier ‘magic’ resources:

- *Vitalvine* (plant)

- *Vitalite Crystal* (rock)
- *Enchantwing* (bug)
- *Mycolumin Essence* (mushroom)

Health Potions are crafted from combinations of a **base magic** resource and **modifying fruit** resource. The base magic resource chooses the theme, and the modifying fruit resource chooses one of the four themed types.

1. Base/Theme		2. Modifier/Type			
		Vitalvine	Vitalite Crystal	Enchantwing	Mycolumin Essence
Heartwood	Harmonics:	<i>Vitalwood Violin</i>	<i>Harmoni-crystal</i>	<i>Chitin Chime</i>	<i>Mycolumin Lyre</i>
Moonstone	Masonry:	<i>Stonelace Ivy</i>	<i>Seismi-crystal</i>	<i>Keystone Relic</i>	<i>Glowgrain Marble</i>
Dark Silk	Secretspinning:	<i>Vitalink Codex</i>	<i>Mysti-crystal</i>	<i>Shadowscript Scroll</i>	<i>Mycolumin Manuscript</i>
Fungus Fiber	Mycofibercraft:	<i>Lifeweave Vest</i>	<i>Prisma-crystal</i>	<i>Enchanted Shadowcloak</i>	<i>Glowthread Garment</i>

The four Gift corresponding activity themes are currently:

- **Music (“Hearwood Harmonics”).** Favored by the Growth creatures. Could later intersect with music “events” (bring instruments to a type of “hangout”), or resource spawning rates or conditions. Ex: Plants could grow well on plots where creatures play certain types of music.
- **Building (“Moonstone Masonry”).** Favored by the Stability creatures. Items were intentionally designed to serve as plausible “gift” items as well as bridge building items or materials.
- **Reading/Writing (“Secretspinning.”)** Favored by the Shadow creatures. Items were intentionally designed to serve as plausible “gift” items and connect to “knowledge” like stories, information, recipes, or data.
- **Fashion/Textiles/Craft (“(Myco-)Fibercraft”)** (i.e., Fashion/Clothing/Crafting). Favored by the Light creatures. Items were intentionally designed to serve as plausible “gift” items and connect to possible decoration, avatar customization, or faction-friendship signaling clothing items (e.g. could boost social stat for interacting with creatures of that faction while wearing). Items use “Fungus Fiber” to craft fabric materials, which could be extended to allow colors by creating colored variants of the “Mycolumin Essence” resource.

INTENTIONALLY OPENING POSSIBLE CONNECTION: Crystal Lens Items. Each theme has a “Crystal” type item made by combining the base resource for that theme with the Vitalite

crystal. These “Crystals”, or combinations of them, can serve as “lenses” allowing players to build “sensors” to collect certain kinds of data (as suggested by the Crystal names).

- **Harmoniccrystal.** Enjoyed as a gift by music-loving creatures to help them tune their instruments. Also used by players to upgrade/craft/create “sensors” to detect “balances”, e.g. in “patch health.”
- **Seismiccrystal.** Enjoyed as a gift by masonry/rock sculpture-loving creatures to help them detect instabilities in their structures (or some such thing, doesn’t matter). Also used by players to upgrade/craft/create “sensors” to detect “imbalances,” monitor stability energy, and/or plot variables like water level.
- **Mysticcrystal.** Enjoyed as a gift by knowledge/reading/books/secrets-loving creatures to help them inspect or predict things (magnifying glass or crystal ball or some such thing, doesn’t matter). Also used by players to upgrade/craft/create “sensors” to monitor energies, “patch health”, creature population or faction health, or other variables from afar.
- **Prismacrystal.** Enjoyed as a gift by fashion/fibercraft-loving creatures to help them add colors to their fabrics. Also used by players to upgrade/craft/create “sensors” to monitor light and color-related variables from afar.

Growth Potions [now Patch Crystals]

There are 16 Growth Potions belonging to 4 Themes: **Growth, Stability, Shadow, Light.**

Growth Potions are crafted from combinations of a **base material** resource and a **modifying sweet** resource. The base material chooses the theme, and the modifying sweet resource chooses one of the four themed item types.

Base ‘material’ resources:

- *Heartwood* (plant)
- *Moonstone* (rock)
- *Dark Silk* (bug)
- *Fungus Fiber* (mushroom)

Modifier sweet resources:

- *Nectar* (plant)
- *Sugar Crystals* (rock)
- *Honeycomb* (bug)
- *Powdered Sugar* (mushroom)

3. Base/Theme		4. Modifier/Type			
		Nectar	Sugar Crystals	Honeycomb	Powdered Sugar
Heartwood	Growth:	<i>Growth Specific</i>	<i>Growth Balanced</i>	<i>Growth Neutral</i>	<i>Growth Synergistic</i>

		<i>Sap Bloom Crystal</i>	<i>Sweet Blossom Crystal</i>	<i>Honey Sprout Crystal</i>	<i>Candy Vine Crystal</i>
Moonstone	Stability:	<i>Stability Balanced</i> <i>Luminaector Stone</i>	<i>Stability Specific</i> <i>Silver Glaze Stone</i>	<i>Stability Synergistic</i> <i>Celestial Honey Stone</i>	<i>Stability Neutral</i> <i>Stardust Stone</i>
Dark Silk	Shadow:	<i>Shadow Neutral</i> <i>Midnight Shard</i>	<i>Shadow Synergistic</i> <i>Nocturnal Sucrose Shard</i>	<i>Shadow Specific</i> <i>Honeyweb Shard</i>	<i>Shadow Balanced</i> <i>Shadowdust Shard</i>
Fungus Fiber	Light:	<i>Light Synergistic</i> <i>Mushloom Gem</i>	<i>Light Neutral</i> <i>Sporeweb Gem</i>	<i>Light Balanced</i> <i>Honeyfung Gem</i>	<i>Light Specific</i> <i>Fungal Fluff Gem</i>

4c. Gift Items Descriptions by Theme

The below is junk/placeholder text that I used GPT to turn notes into bullet points. These are narrative descriptions, not technical. I can edit and fix up later if needed.

Harmonics. All “Harmonics” items are crafted using a Heartwood base, and a modifying magic resource of a different spawn type (plant, mushroom, bug or rock).

- **Vitalwood Violin:**
 - *Base:* Heartwood - given its stability and resonance, perfect for musical instruments.
 - *Mod:* Vitalvine - to enhance the sound and perhaps give it a magical resonance.
- **Mycolumin Lyre:**
 - *Base:* Heartwood - for the body of the lyre.
 - *Mod:* Mycolumin Essence - to make the strings or the body of the lyre glow when played.
- **Chitin Chime:**
 - *Base:* Dark Silk - to hang the chimes.
 - *Mod:* Enchantwings - chitin from the wings can be the actual chimes, giving a unique sound when wind flows.

- **Harmoniccrystal:**
 - *Base:* Moonstone - due to its natural reflective and refractive properties.
 - *Mod:* Vitalite Crystal - to amplify or harmonize with energies around it.

Masonry. All “Masonry” items are crafted using a Moonstone base, and a modifying magic resource of a different spawn type (plant, mushroom, bug or rock).

- **Stonelace Ivy:**
 - *Base:* Moonstone - finely ground into a paste or malleable form.
 - *Mod:* Vitalvine - binding the stone, making it flexible and giving a lacy appearance
- **Glowgrain Marble:**
 - *Base:* Moonstone - primary component of the marble.
 - *Mod:* Mycolumin Essence - giving the marble its luminescence.
- **Keystone Relic:**
 - *Base:* Heartwood - forming the base or body of the relic.
 - *Mod:* Enchantwings - maybe inlaid or adorned on the relic for decoration and magical properties.
- **Seismicrystal:**
 - *Base:* Moonstone - for its durability and natural resonance with earth energies.
 - *Mod:* Vitalite Crystal - allowing the user to sense seismic energies or vibrations.

Secretspinning. All “Secretspinning” items are crafted using a Dark Silk base, and a modifying magic resource of a different spawn type (plant, mushroom, bug or rock).

- **Vitalink Codex:**
 - *Base:* Dark Silk - for binding and pages.
 - *Mod:* Vitalvine - ink or glyphs might be infused with its essence for secret inscriptions.
- **Mycolumin Manuscript:**
 - *Base:* Dark Silk - forming the base of the scroll.
 - *Mod:* Mycolumin Essence - for paintings or illustrations on the scroll.
- **Shadowscript Scroll:**
 - *Base:* Dark Silk - base material of the scroll.
 - *Mod:* Enchantwings - the scroll might shimmer or have patterns from the wings, revealing secrets under specific conditions.
- **Mysticrystal:**
 - *Base:* Moonstone - primary lens component.
 - *Mod:* Enchantwings - enhancing the lens to see hidden magical patterns or auras.

Mycofibercraft. All “Mycofibercraft” items are crafted using a Fungus Fiber base, and a modifying magic resource of a different spawn type (plant, mushroom, bug or rock).

- **Lifeweave Vest:**
 - *Base:* Fungus Fiber - main material of the vest, making it durable.
 - *Mod:* Vitalvine - giving it life-enhancing or healing properties.
- **Glowthread Garment:**

- *Base*: Fungus Fiber - base material.
- *Mod*: Mycolumin Essence - making threads of the garment glow.
- **Enchanted Shadowcloak:**
 - *Base*: Fungus Fiber - main material of the cloak.
 - *Mod*: Enchantwings - for decoration and to possibly give stealth or camouflaging abilities.
- **Prismacrystal:**
 - *Base*: Fungus Fiber.
 - *Mod*: Vitalite Crystal.
 - *Uses*: could allow players to see light levels, Light energy levels, colors, access colors, interact with some later color-based mechanic, or to craft/enable a “watcher” on Light energy levels.

4d. Crafting Logic: Item Categories, Themes and Types

Determining Item Category by Resource Categories

Every recipe combines exactly two resources. As currently specified, the ordering of the two resources does not matter. However, these resources can be thought of as a primary, "base" resource, and a secondary, "modifier" resource.

The combination of base and modifier resource categories (fruit, sweet, magic, material) determine the category of item (food, health potion, growth potion, or gift). Ex: combining one of 4 Fruit resources with one of 4 Sweet resources crafts one of the 16 Food items.

Item Categories from Resource Category Combinations

- **Foods**: Sweet (base) + Fruit (modifier)
- **Health Potions**: Magic (base) + Fruit (modifier)
- **Growth Potions**: Material (base) + Sweet (modifier)
- **Gifts**: Material (base) + Magic (modifier)

Determining Item Theme and Type by Modifier and Base Resource Factions

After the item category is determined, to select which of the 16 items are crafted:

- The spawn behavior of the base resource determines the item/recipe theme.
- The spawn behavior of the modifier resource determines the item type.

This can be thought of as the first (base) ingredient selecting the row in a 4x4 matrix, while the second (mod) ingredient selects which column. The 16 cells each have one of the 16 potions.

This combination logic is illustrated in the recipe matrices in the above section. Item themes and types are also defined in the [Items List](#).

Determining Item Faction and Faction Blend by Modifier and Base Resource Factions

Crafted Item factions take the faction of their base resource. Items also have a "faction blend" determined by the factions of the ingredient resources. Currently, this matters mostly for Growth Potion effects, but could be used to modify effects of foods or potions on creatures. The logic for determining this blend is:

1. **Specific:** If both resources are from the same faction, the item blend is "Faction Specific".
2. **Synergistic:** If the factions of the resources are known to work well together (Light with Growth, or Shadow with Stability), the item blend is "Synergistic".
3. **Balanced:** If the factions of the resources are natural opposites (Light with Shadow, or Growth with Stability), the item blend is "Balanced".
4. **Neutral:** If the factions of the resources don't fall into any of the above categories (Light with Stability, or Shadow with Growth), the item blend is "Mixed".

Determining Item Value and Quality by Resource Values and Qualities

The two ingredient resources each have a value and a quality. The value is the same as the cost to the patch_health to forage that resource. The quality is effectively the fraction of resource value transferred to the item:

Crafted Item Value. Item Value = Value transferred from Resource 1 + Value transferred from Resource 2

$$= .01 * \text{resource1_quality} * \text{resource1_value} + .01 * \text{resource2_quality} * \text{resource2_value}$$

For example, a player could craft an item from:

- Resource1 with a value of 30 and quality of 100%
- Resource2 with a value of 20 and quality of 50%.

The crafted item would have a value of $30 + 10 = 40$.

Crafted Item Quality. This is not necessary, but would make sense to do. An item could end up with a varying "quality" depending on, for example, any of:

- choosing the correct temperature while crafting the item recipe (expanding the "green zone" to three values: high, medium, low), which would require multiple crafting attempts and looking at quality outcomes data
- performance in the crafting interactive (staying close within a target "green zone" temp)
- Leveling up or otherwise getting some sort of achievement or upgrade in the crafting context (has the "Potioneer Level 2" badge, or a Golden Stirring Rod, or w/e)

The item quality would similarly represent the “effectiveness” of the item, when given to a creature. A 50% quality item would have 50% the effect on the creature’s stats.

5. Item Effects

5a. Item Effectiveness, Item Effect

This section outlines the logic behind item effects data and progression in [this document](#).

When a creature consumes an item, a few things are calculated:

- **Item Effectiveness/Response:** How effective is this item for this creature?
- **Item Effects:** How should each of the creature's stats (health, mood, social) or state be updated?

Calculating Item Effectiveness/Response

"Item Effectiveness" is a calculated value reflecting the "fit" of an item to any particular creature, or how well the creature responds to it. Items with the highest Effectiveness for that creature represent e.g. their favorite food, favorite gift, while the lowest Item Effectiveness is the worst possible fit for that creature (e.g. least favorite food, items they hate, etc).

Representing Item Effectiveness. When a player gives an item to a creature, the Item Effectiveness is what is mapped to a visual effect or indicator (e.g. smiley or frown face). However, this mapping can vary depending on player level. For example, a Level 0/Beginner player may see the Item Effectiveness mapped to a categorical variable with only two values: "LIKE" or "DISLIKE." If Item Effectiveness > 50, "LIKE"; if Item Effectiveness < 50, "DISLIKE." At Level 1, the mapping is to values "HATE", "DISLIKE", "LIKE", "LOVE".

Each creature has a ranked list of items in each item category (Foods, Gifts, Health Potions). The top ranked item is its "favorite" and can have a boosted, 2x effect, while the lowest ranked item can have negative effects and/or apply a state to the creature. For each creature, their rankings are then mapped to values of Item Effectiveness with this distribution:

0: 200%
1-3: 100%
4-7: 65%
8-11: 35%
12-14: 0%
15: -100%

Calculating Item Effects

“**Item Effects**” are calculated based on the Item Effectiveness, Item Value, Item Quality, and Item Category. Item Value, Quality and Category are properties of the item, while Item Effectiveness depends on both the Item and Creature it is given to.

- 1) Calculate **Item Max Effect** from **Item Value** and **Item Quality**.
 - **Item Value** was calculated when the item was crafted, and added the values of the resource ingredients. This represents the maximum total effect that an item could have on a creature’s stats.
 - **Item Quality** reflects any changes due to crafting process and/or player level. If a player “levels up” in crafting some area (now has an Advanced Cookbook: Cupcakes) they might craft higher quality cupcakes. Lower quality decreases the effect of items on creatures.
 - **Item Max Effect** = Item Value * Item Quality
- 2) Calculate **Item Total Effect** from **Item Max Effect** and **Item Effectiveness**. The Item Max Effect is a property of the Item itself. The effectiveness depends on its “fit” with the receiving creature. Lower fit means lower total effect.
 - **Item Total Effect** = Item Max Effect * Item Effectiveness
- 3) Distribute **Item Total Effect** across creature stats (health, mood, social) depending on item category.
 - **Item Category Base Effects** distribute item effects across creature stats depending on the item category (Food, Health Potion, Gift). See [Item Base Effects](#) in Appendix for how item categories affect creature stats.
 - **Health Potions** have the greatest effect on creature health, and much smaller effects on creature mood or social stats.
 - **Gifts** have no effect on creature health, but impact creature mood and social (with a slightly larger effect on social).
 - **Foods** impact all stats roughly equally, though have the largest effect on mood.
 - **Growth Potions** do nothing to creatures, and can not be gifted or applied to them.
 - Effects on each individual creature stat are then:
 - Health Effect = Item Total Effect * base_effects[item_category][“health”]
 - Mood Effect = Item Total Effect * base_effects[item_category][“mood”]
 - Social Effect = Item Total Effect * base_effects[item_category][“social”]

Calculating Item Rankings

A creature’s item rankings depend on their faction and style, and the base and mod resource types of each item. Each faction has an order in which they like base and mod resource types.

The effect of ranking by resource types is that the creatures have preferences for different item themes (e.g. Growth likes Smoothies and Cupcakes best; Stability likes Jelly and Pastry) as well as “flavors” within each (e.g. Growth likes Peach and Tropical). However, instead of specifying preferences by Theme directly, preferences by resource type allows the preferences to apply across all categories of item (Foods, Gifts and Health Potions). This way, Data collected about

Foods could be re-used to predict Gifts preferences.

```
BASE_ORDER_BY_FACTION = {
    "Growth": ["Plant", "Mushroom", "Bug", "Rock"],
    "Stability": ["Rock", "Bug", "Mushroom", "Plant"],
    "Shadow": ["Bug", "Rock", "Plant", "Mushroom"],
    "Light": ["Mushroom", "Plant", "Rock", "Bug"]
}
```

```
MOD_ORDER_BY_FACTION = {
    "Growth": ["Rock", "Mushroom", "Bug", "Plant"],
    "Stability": ["Plant", "Bug", "Mushroom", "Rock"],
    "Shadow": ["Mushroom", "Rock", "Plant", "Bug"],
    "Light": ["Bug", "Plant", "Rock", "Mushroom"]
}
```

There are three components to the ranking:

- 1) If the faction likes the base resource type (type of item)
- 2) If the faction likes the mod resource type (flavor/variety of item)
- 3) The style preferences for specific base, mod combinations (just used to create variation in preferences by style)

Style E creatures behave like Style A of the Opposite faction (e.g. Growth Style E behaves like Stability Style A).

```
effective_faction = FACTION_OPPOSITES[faction] if style == "E" else faction
```

The actual ranking logic then works in the following way, which calculates a unique rank and score for each of the 16 items, for each creature faction and style:

- 1) Each item starts at a score of 0.
- 2) If the base resource type is in the Top 2 for that creature's (effective) faction, add 8. This creates the major division of item scores, between "LIKE" (+8) and "DISLIKE" (+0).
- 3) If the mod resource type is in the Top 2 for that creature's (effective) faction, add 4. This further segments the item scores into four levels: "LOVE" (+8 and +4), "LIKE" (+8), "DISLIKE" (+4), and "HATE" (+0). There are four items at each level; for example the items in the current highest level (score of 12) are the ones in the "top 2" base AND "top 2" mod ingredients.
- 4) A final "style score" from 0-3 is added to each item score. First a "combo_score" from 0 to 3 is given to each item depending on its combined ranking of base and mod ingredients. Then, those scores are shifted depending on the creature style. This now gives each item a unique score, and rank, from 0 to 15, and gives each creature style a unique order of preferences:

```
def calculate_style_score(base_rank, mod_rank, style):
    combo_score = 2 * (base_rank % 2) + (mod_rank % 2)
    style_shift = "ABCDE".index(style) % 4
    style_combo = (combo_score + style_shift) % 4
    return style_combo
```

Note that Style E now has the preference order of a Style A.

Potion Effects on Creature States

In addition to effects on creature stats, potions can affect (remove) creature states, currently: Allergy, Anomaly, Sleeping Sickness, Flu. When these states are applied, the creature's health or mood will decrease daily until the state is removed. To remove the state, and "cure" the creature's sickness, does not mean to simply to increase their health stat– creatures can not be "cured" by feeding them lots of treats. Instead, sicknesses require specific potion cures.

Each of the 4 factions can require a specific cure for each of the 4 sicknesses (I haven't thought about this too much yet, but here's a first pass):

- Flu is cured by Teas, with each faction requiring its own flavor.
- Allergy is cured by potions where both ingredients are associated with the faction.
- Sleeping Sickness is cured by bobas with flavors of the synergistic faction.
- Anomalies are cured by tonics of the opposite faction.

	3. Modifier/Type			
	<i>Allergy (faction-specific)</i>	<i>Flu (same faction teas)</i>	<i>Sleeping Sickness (synergistic boba)</i>	<i>Anomaly (opposite faction tonic)</i>
Growth	<i>Wheatberry Green Tea</i>	<i>Wheatberry Green Tea</i>	<i>Sporefruit Sparkle Boba</i>	<i>Stonefruit Shield Syrup</i>
Stability	<i>Stonefruit Shield Syrup</i>	<i>Peach Oolong Tea</i>	<i>Cocoberry Bliss Boba</i>	<i>Wheatberry Guard Tonic</i>
Shadow	<i>Cocoa Mystic Brew</i>	<i>Cocoa Mint Tea</i>	<i>Stonefruit Crystal Boba</i>	<i>Sporefruit Defense Tonic</i>
Light	<i>Sporefruit Sparkle Boba</i>	<i>Tropical Tea</i>	<i>Wheatberry Glow Boba</i>	<i>Cocoa Barrier Blend</i>

6. Growth Potions and Island Health

Patch Health. The state of a Patch is defined by its `patch_health`. The health of a patch is subdivided into four types, one for each faction. This can be thought of as available "faction nutrients" at each patch location. If a faction nutrient is low, then resources of that faction (requiring that nutrient) spawn less often or not at all.

```
patch_health: {  
  
    "Growth": 500  
  
    "Stability": 1000  
  
    "Shadow": 50  
  
    "Light": 250  
  
}
```

Decreases to patch health. Patch health decreases when resources are foraged from that patch by a player, with the decrease applying to the faction of the resource. Foraging "Light" resources decreases the "Light" component of patch health, by the "value" of the resource (essentially, how much it depletes the patch health to forage it). Resource values are specified in the [Resources List](#).

Increases to patch health.

- Creatures increase patch health during their `tend` pastime, transferring creature stat points into patch health points.
- Players can increase or alter `patch_health` by applying growth potions. The 16 growth potion recipes each have a unique combination of: base faction (Growth, Stability, Shadow, Light) and a faction blend (Specific, Balanced, Synergistic, Neutral).
- The total points increase to patch health is specified by the `value` of the growth potion, while the way it is distributed across the four faction values depends on the potion type (Specific, Synergistic, Balanced or Neutral). See [Growth Potion Types](#).
- There are many possible things to do here, but one straightforward way this could work:
 - **Specific Potions:** add all potion value to the base faction. Ex: a Light Specific Potion with value 100 added to the example patch above would turn the Light number to 350, leaving all others the same.
 - **Balanced Potions:** add value to and equalize the two factions in the blend. Ex: a Light Balanced Potion with value 100 added to the example patch above would turn Light and Shadow numbers both to 200.

- **Synergistic Potions:** add value and swap between the two synergistic factions. Ex: a Light Synergistic Potion with value 100 added to the example patch above would turn Light to 350, then swap with Growth, ending at Light: 500, Growth: 350.
- **Neutral Potions:** could add to the base faction then randomize energies, add its value to a random energy type, add equal amounts to each energy type, or add to the total then equalize all energies. Ex: a Light Balanced Potion with value 100 added to the example patch above would turn all values to 475.

Update Logic

All tentative, TBD. The following are updates to the game state variables, either triggered by some event or to happen at a certain time interval (most are daily):

- Plot “water level” decreases by some amount per Dry day. Could be a different decrease per day for different islands or plots.
- Player-creature relationship level drops by 1 point every day. Just interacting with a friend creature should increase it 1 point– i.e. “maintaining friendships.”
- Creature stats might update overnight, ex: when low health, mood also drops each day.
- Applied states can update creature stats (e.g. health decreases by 2 points every day the creature has “Flu” or “Anomaly” applied).
- Patch health can update during ‘tend’ by an amount that depends on creature stats. If creatures have high stats, and are ‘tending’ a plot with low stats, they should potentially transfer more.

Notes on Connections to Progressions

Shops

- Creatures request an Item category or Resource Category. Specific item that would best fit the request is not known.
- Creature requests follow activity preference patterns.
- Questions: what is shop success metric? Consequences for neglecting the shop requests (poor mood, like Medical Building)?

Bridges

- Factions have a `player_friendships_threshold` to allow you to build a bridge.
- The “masonry” themed items might be used for bridge repair.
- Bridge repair makes travel easier for the player.
- Bridge repair can make the probabilities of creatures traveling to different islands higher, shifting the distribution of resources on the islands. This can be part of “restoration” of the island habitats as well as “rebuilding community.”

Anomalies

To be written. Anomalies apply states to creatures and affect stats. They have to be removed by a specific Health potion (general rules do not apply). The easiest way to do these “Potions Cure” rules may be to take the 4x4 POTIONS_RECIPES grid, and select row by creature faction, and column by some kind of symptom/sickness or creature “state” to remove.

Anomalies can affect creature stats and state: its health (and/or mood) stats can be decreased, and a state applied (e.g. "Cough", "Sick", "Anomaly Type A"). Applied states can update creature stats (e.g. health decreases by 2 points every day the creature has “Cough” applied), or can alter behavior patterns (if “Anomaly A” applied, creature only chooses Activity, or now prefers the opposite island). These will be specified in a section on [Anomalies](#).

Imbalance Events

Triggered by unbalanced gameplay and faction or island stat imbalances. What counts as an “imbalance” is TBD, but it will most likely be a function of the two factions’ relative 3-stat averages.

Ex: if (Light(health+social+mood) - Shadow(health+social+mood)) / TOTALS > THRESHOLD

Creature “Hangouts”

- When creatures “hangout” at a plot and/or do an activity—achievement for delivering correct food or gifts to each attendee.
- Attendee list advertised in advance in the **Newspaper**.

Newspaper

- Weather, Weather Predictor
- Faction stats, averages (mapped to emoji states), and/or state “counts” (e.g. “5 Light creatures have a Cough.”)
- Island stats
- Creature Behaviors info
- Need pickaxe to mine rocks

Misc

- Need bug net to capture bugs
- Need watering can to craft growth potions
- Find a lot of a common resource (know which island)
- Find a few of a rare resource (know plot)
- 100% quality of X (know to go off-island for higher quality)
- Growth potion is the most direct route to solve problems.
- Refinery/Upcycle (1 + MANY --> boost quality out)

Appendix

Weather System

```
"RAIN_CONDITIONS" = ["Rainy", "Dry"]

"SUN_CONDITIONS" = ["Sunny", "Cloudy"]

"SEASONAL_WEATHERS" = {

    "Spring": {

        "rain_chances": {"Rainy": 0.7, "Dry": 0.3},

        "sun_chances": {"Cloudy": 0.3, "Sunny": 0.7}

    },

    "Summer": {

        "rain_chances": {"Rainy": 0.3, "Dry": 0.7},

        "sun_chances": {"Cloudy": 0.3, "Sunny": 0.7}

    },

    "Fall": {

        "rain_chances": {"Rainy": 0.3, "Dry": 0.7},

        "sun_chances": {"Cloudy": 0.7, "Sunny": 0.3}

    },

    "Winter": {

        "rain_chances": {"Rainy": 0.7, "Dry": 0.3},

        "sun_chances": {"Cloudy": 0.7, "Sunny": 0.3}

    }

}

import random
from config.config import (ISLAND_NAMES, RAIN_CONDITIONS,
SUN_CONDITIONS, SEASONAL_WEATHERS)

class WeatherSystem:
    def __init__(self, clock):
```

```

self.clock = clock
self.season = clock.get_season()
self.island_weathers = {island: {
    "sun_condition": random.choice(SUN_CONDITIONS),
    "rain_condition": random.choice(RAIN_CONDITIONS)
} for island in ISLAND_NAMES}
self.data = []
self.next()

def get_island_weathers(self):
    return self.island_weathers

def get_data(self):
    return self.data

def get_weather_on_island(self, island):
    return self.island_weathers[island]

def log_weather(self):
    """Logs the weather for each island for the current day."""
    for island_name in ISLAND_NAMES:
        self.data.append({
            "day": self.clock.get_day(),
            "island_name": island_name,
            "sun_condition":
self.island_weathers[island_name]["sun_condition"],
            "rain_condition":
self.island_weathers[island_name]["rain_condition"]
        })

def next(self):
    """Updates the weather conditions on each island depending on
season."""
    self.season = self.clock.get_season()
    sun_chances = SEASONAL_WEATHERS[self.season]["sun_chances"]
    rain_chances = SEASONAL_WEATHERS[self.season]["rain_chances"]
    sun_condition = random.choice(list(sun_chances.keys())),
weights=list(sun_chances.values()))
    rain_condition = random.choice(list(rain_chances.keys())),
weights=list(rain_chances.values()))

    # TODO: Update this section if per-island weathers based on
wind is implemented in the future.
    for island in ISLAND_NAMES:

```

```
        self.island_weathers[island]["sun_condition"] =
sun_condition
        self.island_weathers[island]["rain_condition"] =
rain_condition

    self.log_weather()
```

Creatures List

"Light"

```
{"creature_name": "Glint", "creature_color": "Blue", "creature_style": "A"},
{"creature_name": "Prismaros", "creature_color": "Blue", "creature_style": "B"},
{"creature_name": "Aurora", "creature_color": "Blue", "creature_style": "C"},
{"creature_name": "Celestia", "creature_color": "Blue", "creature_style": "D"},
{"creature_name": "Radiant", "creature_color": "Blue", "creature_style": "E"},
{"creature_name": "Amethyst", "creature_color": "Purple", "creature_style": "A"},
{"creature_name": "Luminara", "creature_color": "Purple", "creature_style": "B"},
{"creature_name": "Iris", "creature_color": "Purple", "creature_style": "C"},
{"creature_name": "Seraphim", "creature_color": "Purple", "creature_style": "D"},
{"creature_name": "Violet", "creature_color": "Purple", "creature_style": "E"},
{"creature_name": "Pyroshimmer", "creature_color": "Red", "creature_style": "A"},
{"creature_name": "Blaze", "creature_color": "Red", "creature_style": "B"},
{"creature_name": "Radiance", "creature_color": "Red", "creature_style": "C"},
{"creature_name": "Ignatia", "creature_color": "Red", "creature_style": "D"},
{"creature_name": "Phoenix", "creature_color": "Red", "creature_style": "E"},
{"creature_name": "Viridium", "creature_color": "Green", "creature_style": "A"},
{"creature_name": "Meadowlark", "creature_color": "Green", "creature_style": "B"},
{"creature_name": "Lumigrove", "creature_color": "Green", "creature_style": "C"},
{"creature_name": "Verdant", "creature_color": "Green", "creature_style": "D"},
{"creature_name": "Zephyr", "creature_color": "Green", "creature_style": "E"},
```

```
{"creature_name": "Solara", "creature_color": "Orange", "creature_style": "A"},  
{"creature_name": "Lumos", "creature_color": "Orange", "creature_style": "B"},  
{"creature_name": "Aurelia", "creature_color": "Orange", "creature_style": "C"},  
{"creature_name": "Luminary", "creature_color": "Orange", "creature_style": "D"},  
{"creature_name": "Radiantwing", "creature_color": "Orange", "creature_style": "E"}],
```

"Shadow"

```
{"creature_name": "Umbra", "creature_color": "Blue", "creature_style": "A"},  
{"creature_name": "Shade", "creature_color": "Blue", "creature_style": "B"},  
{"creature_name": "Duskray", "creature_color": "Blue", "creature_style": "C"},  
{"creature_name": "Twilight", "creature_color": "Blue", "creature_style": "D"},  
{"creature_name": "Nyx", "creature_color": "Blue", "creature_style": "E"},  
{"creature_name": "Obsidian", "creature_color": "Purple", "creature_style": "A"},  
{"creature_name": "Shadowstrike", "creature_color": "Purple", "creature_style": "B"},  
{"creature_name": "Eclipse", "creature_color": "Purple", "creature_style": "C"},  
{"creature_name": "Nocturna", "creature_color": "Purple", "creature_style": "D"},  
{"creature_name": "Nox", "creature_color": "Purple", "creature_style": "E"},  
{"creature_name": "Pyroshadow", "creature_color": "Red", "creature_style": "A"},  
{"creature_name": "Ember", "creature_color": "Red", "creature_style": "B"},  
{"creature_name": "Shadowfire", "creature_color": "Red", "creature_style": "C"},  
{"creature_name": "Inferno", "creature_color": "Red", "creature_style": "D"},  
{"creature_name": "Ashen", "creature_color": "Red", "creature_style": "E"},  
{"creature_name": "Verdigris", "creature_color": "Green", "creature_style": "A"},  
{"creature_name": "Murk", "creature_color": "Green", "creature_style": "B"},  
{"creature_name": "Shadowthorn", "creature_color": "Green", "creature_style": "C"},  
{"creature_name": "Mossbane", "creature_color": "Green", "creature_style": "D"},  
{"creature_name": "Shadowleaf", "creature_color": "Green", "creature_style": "E"},  
{"creature_name": "Umbertide", "creature_color": "Orange", "creature_style": "A"},  
{"creature_name": "Gloomsong", "creature_color": "Orange", "creature_style": "B"},
```

```
{ "creature_name": "Shadowflare", "creature_color": "Orange", "creature_style": "C"},  
{ "creature_name": "Twilightspire", "creature_color": "Orange", "creature_style": "D"},  
{ "creature_name": "Nightshade", "creature_color": "Orange", "creature_style": "E"}],
```

"Growth"

```
{ "creature_name": "Flora", "creature_color": "Blue", "creature_style": "A"},  
{ "creature_name": "Lush", "creature_color": "Blue", "creature_style": "B"},  
{ "creature_name": "Sylvan", "creature_color": "Blue", "creature_style": "C"},  
{ "creature_name": "Meadow", "creature_color": "Blue", "creature_style": "D"},  
{ "creature_name": "Zephyrus", "creature_color": "Blue", "creature_style": "E"},  
{ "creature_name": "Blossom", "creature_color": "Purple", "creature_style": "A"},  
{ "creature_name": "Petal", "creature_color": "Purple", "creature_style": "B"},  
{ "creature_name": "Thorn", "creature_color": "Purple", "creature_style": "C"},  
{ "creature_name": "Rose", "creature_color": "Purple", "creature_style": "D"},  
{ "creature_name": "Fern", "creature_color": "Purple", "creature_style": "E"},  
{ "creature_name": "Mosswhisper", "creature_color": "Red", "creature_style": "A"},  
{ "creature_name": "Bramble", "creature_color": "Red", "creature_style": "B"},  
{ "creature_name": "Sunleaf", "creature_color": "Red", "creature_style": "C"},  
{ "creature_name": "Emberbloom", "creature_color": "Red", "creature_style": "D"},  
{ "creature_name": "Chlorophyll", "creature_color": "Red", "creature_style": "E"},  
{ "creature_name": "Viridius", "creature_color": "Green", "creature_style": "A"},  
{ "creature_name": "Willow", "creature_color": "Green", "creature_style": "B"},  
{ "creature_name": "Oakheart", "creature_color": "Green", "creature_style": "C"},  
{ "creature_name": "Ivy", "creature_color": "Green", "creature_style": "D"},  
{ "creature_name": "Thicket", "creature_color": "Green", "creature_style": "E"},  
{ "creature_name": "Solstice", "creature_color": "Orange", "creature_style": "A"},  
{ "creature_name": "Gleam", "creature_color": "Orange", "creature_style": "B"},  
{ "creature_name": "Radiant Blossom", "creature_color": "Orange", "creature_style":  
"C"},
```

```
{"creature_name": "Vivaro", "creature_color": "Orange", "creature_style": "D"},  
{"creature_name": "Evergreen", "creature_color": "Orange", "creature_style": "E"}],
```

"Stability"

```
{"creature_name": "Granite", "creature_color": "Blue", "creature_style": "A"},  
{"creature_name": "Steadfast", "creature_color": "Blue", "creature_style": "B"},  
{"creature_name": "Vanguard", "creature_color": "Blue", "creature_style": "C"},  
{"creature_name": "Shieldstone", "creature_color": "Blue", "creature_style": "D"},  
{"creature_name": "Ironclad", "creature_color": "Blue", "creature_style": "E"},  
{"creature_name": "Ash", "creature_color": "Purple", "creature_style": "A"},  
{"creature_name": "Stonewall", "creature_color": "Purple", "creature_style": "B"},  
{"creature_name": "Firmament", "creature_color": "Purple", "creature_style": "C"},  
{"creature_name": "Grit", "creature_color": "Purple", "creature_style": "D"},  
{"creature_name": "Solidarity", "creature_color": "Purple", "creature_style": "E"},  
{"creature_name": "Ambrose", "creature_color": "Red", "creature_style": "A"},  
{"creature_name": "Hearth", "creature_color": "Red", "creature_style": "B"},  
{"creature_name": "Ironheart", "creature_color": "Red", "creature_style": "C"},  
{"creature_name": "Resolute", "creature_color": "Red", "creature_style": "D"},  
{"creature_name": "Valiant", "creature_color": "Red", "creature_style": "E"},  
{"creature_name": "Verdantia", "creature_color": "Green", "creature_style": "A"},  
{"creature_name": "Stalwart", "creature_color": "Green", "creature_style": "B"},  
{"creature_name": "Strongroot", "creature_color": "Green", "creature_style": "C"},  
{"creature_name": "Solidrock", "creature_color": "Green", "creature_style": "D"},  
{"creature_name": "Endurance", "creature_color": "Green", "creature_style": "E"},  
{"creature_name": "Aurum", "creature_color": "Orange", "creature_style": "A"},  
{"creature_name": "Guardian", "creature_color": "Orange", "creature_style": "B"},  
{"creature_name": "Ironcliff", "creature_color": "Orange", "creature_style": "C"},  
{"creature_name": "Steelsong", "creature_color": "Orange", "creature_style": "D"},  
{"creature_name": "Lastingheart", "creature_color": "Orange", "creature_style": "E"}
```

```
]
```

Resources List

```
[
```

```
{  
  "resource_name": "Wheatberry",  
  "resource_category": "fruit",  
  "resource_spawn_behavior": "plants",  
  "resource_faction": "Growth",  
  "resource_shade_preference": "PART SUN",  
  "resource_water_preference": "VERY WET",  
  "resource_commonness": 1,  
  "resource_value": 1  
},  
{  
  "resource_name": "Sporefruit",  
  "resource_category": "fruit",  
  "resource_spawn_behavior": "mushrooms",  
  "resource_faction": "Growth",  
  "resource_shade_preference": "PART SUN",  
  "resource_water_preference": "WET",  
  "resource_commonness": 1,  
  "resource_value": 1  
},  
{  
  "resource_name": "Cocoaberry",  
  "resource_category": "fruit",  
  "resource_spawn_behavior": "bugs",  
  "resource_faction": "Growth",
```



```

    "resource_shade_preference": "PART SHADE",

    "resource_water_preference": "DRY",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Stonefruit",

    "resource_category": "fruit",

    "resource_spawn_behavior": "rocks",

    "resource_faction": "Growth",

    "resource_shade_preference": "PART SHADE",

    "resource_water_preference": "VERY DRY",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Nectar",

    "resource_category": "sweet",

    "resource_spawn_behavior": "plants",

    "resource_faction": "Light",

    "resource_shade_preference": "FULL SUN",

    "resource_water_preference": "VERY WET",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Powdered Sugar",

    "resource_category": "sweet",

    "resource_spawn_behavior": "mushrooms",

```

```

    "resource_faction": "Light",
    "resource_shade_preference": "FULL SUN",
    "resource_water_preference": "WET",
    "resource_commonness": 1,
    "resource_value": 1
},
{
    "resource_name": "Honeycomb",
    "resource_category": "sweet",
    "resource_spawn_behavior": "bugs",
    "resource_faction": "Light",
    "resource_shade_preference": "PART SUN",
    "resource_water_preference": "DRY",
    "resource_commonness": 1,
    "resource_value": 1
},
{
    "resource_name": "Sugar Crystals",
    "resource_spawn_behavior": "rocks",
    "resource_category": "sweet",
    "resource_faction": "Light",
    "resource_shade_preference": "PART SUN",
    "resource_water_preference": "VERY DRY",
    "resource_commonness": 1,
    "resource_value": 1
},
{
    "resource_name": "Vitalvine",
    "resource_category": "magic",

```

```

    "resource_spawn_behavior": "plants",

    "resource_faction": "Shadow",

    "resource_shade_preference": "PART SHADE",

    "resource_water_preference": "VERY WET",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Mycolumin Essence",

    "resource_category": "magic",

    "resource_spawn_behavior": "mushrooms",

    "resource_faction": "Shadow",

    "resource_shade_preference": "PART SHADE",

    "resource_water_preference": "WET",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Enchantwings",

    "resource_category": "magic",

    "resource_spawn_behavior": "bugs",

    "resource_faction": "Shadow",

    "resource_shade_preference": "FULL SHADE",

    "resource_water_preference": "DRY",

    "resource_commonness": 1,

    "resource_value": 1
},
{

    "resource_name": "Vitalite Crystal",

```

```

    "resource_category": "magic",
    "resource_spawn_behavior": "rocks",
    "resource_faction": "Shadow",
    "resource_shade_preference": "FULL SHADE",
    "resource_water_preference": "VERY DRY",
    "resource_commonness": 1,
    "resource_value": 1
},
{
    "resource_name": "Heartwood",
    "resource_category": "material",
    "resource_spawn_behavior": "plants",
    "resource_faction": "Stability",
    "resource_shade_preference": "PART SUN",
    "resource_water_preference": "VERY WET",
    "resource_commonness": 1,
    "resource_value": 1
},
{
    "resource_name": "Fungus Fiber",
    "resource_category": "material",
    "resource_spawn_behavior": "mushrooms",
    "resource_faction": "Stability",
    "resource_shade_preference": "PART SUN",
    "resource_water_preference": "WET",
    "resource_commonness": 1,
    "resource_value": 1
},
{

```

```

    "resource_name": "Dark Silk",
    "resource_category": "material",
    "resource_spawn_behavior": "bugs",
    "resource_faction": "Stability",
    "resource_shade_preference": "PART SHADE",
    "resource_water_preference": "DRY",
    "resource_commonness": 1,
    "resource_value": 1
  },
  {
    "resource_name": "Moonstone",
    "resource_category": "material",
    "resource_spawn_behavior": "rocks",
    "resource_faction": "Stability",
    "resource_shade_preference": "PART SHADE",
    "resource_water_preference": "VERY DRY",
    "resource_commonness": 1,
    "resource_value": 1
  }
]

```

Items List

Organized by Item Category > Item Theme > Item Type.

"Food"

```

"Smoothie": ["Berry Burst Smoothie", "Stonefruit Spritzer", "Chocolate
Delight", "Tropical Punch"],

"Jelly": ["Berry Burst Jelly", "Apricot Jelly", "Chocolate Jelly", "Tropical
Fruit Gel"],

```

"Pastry": ["Wheatberry Pastry", "Peach Pie", "Chocolate Tart", "Tropical Fruit Puff"],

"Cupcake": ["Vanilla Cupcake", "Peach Cupcake", "Chocolate Cupcake", "Funfetti Cupcake"]

"Health Potion"

"Rejuvenation Tea": ["Wheatberry Green Tea", "Stonefruit Oolong Tea", "Cocoa Mint Tea", "Sporefruit Herbal Tea"],

"Protection Tonic": ["Wheatberry Guard Tonic", "Stonefruit Shield Syrup", "Cocoa Barrier Blend", "Sporefruit Defense Tonic"],

"Enchantment Brew": ["Wheatberry Luminescent Brew", "Stonefruit Illusion Brew", "Cocoa Mystic Brew", "Sporefruit Enigma Brew"],

"Luminous Boba": ["Wheatberry Glow Boba", "Stonefruit Crystal Boba", "Cocoaberry Bliss Boba", "Sporefruit Sparkle Boba"]

"Gift"

"Harmonics": ["Vitalwood Violin", "Harmonicrystal", "Chitin Chime", "Mycolumin Lyre"],

"Masonry": ["Stonelace Ivy", "Seismicrystal", "Keystone Relic", "Glowgrain Marble"],

"Secretspinning": ["Vitalink Codex", "Mysticrystal", "Shadowscript Scroll", "Mycolumin Manuscript"],

"Mycofibercraft": ["Lifeweave Vest", "Prismacrystal", "Enchanted Shadowcloak", "Glowthread Garment"]

"Growth Potion"

"Growth": ["Growth Specific", "Growth Balanced", "Growth Neutral", "Growth Synergistic"],

"Stability": ["Stability Balanced", "Stability Specific", "Stability Synergistic", "Stability Neutral"],

"Shadow": ["Shadow Neutral", "Shadow Synergistic", "Shadow Specific", "Shadow Balanced"],

```
"Light": ["Light Synergistic", "Light Neutral", "Light Balanced", "Light  
Specific"]
```

Item Base Effects

```
"Health Potion": {  
    "health": 4,  
    "mood": 1,  
    "social": 1  
},  
"Food": {  
    "health": 2,  
    "mood": 2,  
    "social": 2  
},  
"Gift": {  
    "health": 0,  
    "mood": 3,  
    "social": 3  
},  
"Growth Potion": {  
    "health": 0,  
    "mood": 0,  
    "social": 0  
}
```

Creature Behavior Effects

CB1: Pastime and Time of Day

Current probability modifiers for different factions to select particular pastimes at different times of day. Currently they do not do this probabilistically (all creatures of the faction will choose the same pastime, since the values of all others are zero).

```
"AM Early": {
  "Light": { "hangout": 1, "rest": 0, "activity": 0, "tend": 0 },
  "Shadow": { "hangout": 0, "rest": 1, "activity": 0, "tend": 0 },
  "Growth": { "hangout": 0, "rest": 0, "activity": 1, "tend": 0 },
  "Stability": { "hangout": 0, "rest": 0, "activity": 0, "tend": 1 }
},
"AM Late": {
  "Light": { "hangout": 0, "rest": 0, "activity": 0, "tend": 1 },
  "Shadow": { "hangout": 1, "rest": 0, "activity": 0, "tend": 0 },
  "Growth": { "hangout": 0, "rest": 1, "activity": 0, "tend": 0 },
  "Stability": { "hangout": 0, "rest": 0, "activity": 1, "tend": 0 }
},
"PM Early": {
  "Light": { "hangout": 0, "rest": 0, "activity": 1, "tend": 0 },
  "Shadow": { "hangout": 0, "rest": 0, "activity": 0, "tend": 1 },
  "Growth": { "hangout": 1, "rest": 0, "activity": 0, "tend": 0 },
  "Stability": { "hangout": 0, "rest": 1, "activity": 0, "tend": 0 }
},
"PM Late": {
  "Light": { "hangout": 0, "rest": 1, "activity": 0, "tend": 0 },
  "Shadow": { "hangout": 0, "rest": 0, "activity": 1, "tend": 0 },
  "Growth": { "hangout": 0, "rest": 0, "activity": 0, "tend": 1 },
  "Stability": { "hangout": 1, "rest": 0, "activity": 0, "tend": 0 }
}
```

CB1: Island Preferences for Plot Choice (for 'tend' or 'hangout' pastime)

```
"HOME_ISLAND_MULTIPLIER": 6, (6x multiplier for plots on home island)
"SYNERGISTIC_ISLAND_MULTIPLIER": 3, (3x multiplier for plots on synergistic island)
"NEUTRAL_ISLAND_MULTIPLIER": 2, (2x multiplier for plots on neutral island)
```


Plot Variables List

Specifying the number of plots on each island with shade levels FULL SUN, PART SUN, PART SHADE, or FULL SHADE, or water levels VERY DRY, DRY, WET, or VERY WET.

The distributions are shown in the World Map section at the beginning of this document.

EXAMPLE PLOT DATA:

```
"NINE_PLOTS": [
{
  "name": "Small Clearing",
    "pastime_types": {
      "hangout": 1,
      "tend": 1,
      "activity": 0,
      "rest": 1
    }
},
{
  "name": "Secret Forest",
    "pastime_types": {
      "hangout": 0,
      "tend": 1,
      "activity": 1,
      "rest": 1
    }
},
{
  "name": "North Fields",
  "layout": "basic_plot",
    "pastime_types": {
      "hangout": 0,
      "tend": 1,
      "activity": 1,
      "rest": 0
    }
},
{
  "name": "Large Garden",
    "pastime_types": {
```

```

        "hangout": 1,
        "tend": 1,
        "activity": 0,
        "rest": 1
    }
},
{
    "name": "Community Garden",
    "pastime_types": {
        "hangout": 0,
        "tend": 1,
        "activity": 1,
        "rest": 1
    }
},
{
    "name": "South Fields",
    "pastime_types": {
        "hangout": 1,
        "tend": 1,
        "activity": 0,
        "rest": 0
    }
},
{
    "name": "Large Clearing",
    "pastime_types": {
        "hangout": 0,
        "tend": 1,
        "activity": 0,
        "rest": 1
    }
},
{
    "name": "Small Garden",
    "pastime_types": {
        "hangout": 1,
        "tend": 1,
        "activity": 0,
        "rest": 0
    }
},

```

```

{
  "name": "Musical Mines",
  "pastime_types": {
    "hangout": 0,
    "tend": 1,
    "activity": 1,
    "rest": 0
  }
}

```

Spawn Behavior Effects

```

"time_of_day": {
  "resource_spawn_behavior": {
    "AM Early": {
      "bugs": {"weight": 2, "quality": 100}
    },
    "AM Late": {
      "rocks": {"weight": 2, "quality": 100},
    },
    "PM Early": {
      "plants": {"weight": 2, "quality": 100},
    },
    "PM Late": {
      "mushrooms": {"weight": 2, "quality": 100}
    }
  }
},
"rain_condition": {
  "resource_spawn_behavior": {
    "Rainy": {
      "plants": {"weight": 2, "quality": 80}
    },
    "Dry": {
      "rocks": {"weight": 2, "quality": 80}
    }
  }
},
"sun_condition": {
  "resource_spawn_behavior": {
    "Sunny": {

```

```

        "mushrooms": {"weight": 2, "quality": 80}
    },
    "Cloudy": {
        "bugs": {"weight": 2, "quality": 80}
    }
}

},
"plot_shade_level": {
    "resource_shade_preference": {
        "HIGH": {
            "HIGH": {"weight": 3, "quality": 100},
            "MEDIUM": {"weight": 2, "quality": 90},
            "LOW": {"weight": 1, "quality": 80}
        },
        "MEDIUM": {
            "HIGH": {"weight": 1, "quality": 90},
            "MEDIUM": {"weight": 2, "quality": 100},
            "LOW": {"weight": 1, "quality": 90}
        },
        "LOW": {
            "HIGH": {"weight": 1, "quality": 80},
            "MEDIUM": {"weight": 2, "quality": 90},
            "LOW": {"weight": 3, "quality": 100}
        }
    }
},
"plot_water_level": {
    "HIGH": {
        "HIGH": {"weight": 3, "quality": 100},
        "MEDIUM": {"weight": 2, "quality": 90},
        "LOW": {"weight": 1, "quality": 80}
    },
    "MEDIUM": {
        "HIGH": {"weight": 1, "quality": 90},
        "MEDIUM": {"weight": 2, "quality": 100},
        "LOW": {"weight": 1, "quality": 90}
    },
    "LOW": {
        "HIGH": {"weight": 1, "quality": 80},
        "MEDIUM": {"weight": 2, "quality": 90},
        "LOW": {"weight": 3, "quality": 100}
    }
}

```

```

},
"plot_id": {
  "resource_name": {
    "Growth > Secret Forest": {
      "Heartwood": {"weight": 4, "quality": 100}
    },
    "Stability > Secret Forest": {
      "Bedrock": {"weight": 4, "quality": 100}
    },
    "Shadow > Secret Forest": {
      "Dark Silk": {"weight": 4, "quality": 100}
    },
    "Light > Secret Forest": {
      "Fungus Fiber": {"weight": 4, "quality": 100}
    }
  }
}
}

```